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NPTEL (https://swayam.gov.in/explorer?ncCode=NPTEL) » The Joy of Computing using Python (course)

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Course outline

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Week 2 ()

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Week 4 : Assignment 4

The due date for submitting this assignment has passed.

Due on 2025-02-19, 23:59 IST.

Assignment submitted on 2025-02-18, 06:06 IST

1) Which of the following options provides the general formula for the magic constant of a ***1 point*** magic square of size n, where all elements are distinct numbers from 1 to n^2 ?

- [x] $\frac{n(n^2+1)}{2}$
- [ ] $\frac{n^3}{2}$
- [ ] $\frac{n^3+2n}{2}$
- [x] $\frac{n^4+n^2}{2n}$

Yes, the answer is correct.
Score: 1

Accepted Answers:
$\frac{n(n^2+1)}{2}$
$\frac{n^4+n^2}{2n}$

Practice is the key (unit? unit=78&lesson=79)
Magic Square: Hit and Trial 01 (unit? unit=78&lesson=80)
Magic Square: Hit and Trial 02 (unit? unit=78&lesson=81)
Magic Square: Hit and Trial 03 (unit? unit=78&lesson=82)
Magic Square: Hit and Trial 04 (unit? unit=78&lesson=83)
Magic Square: Hit and Trial 05 (unit? unit=78&lesson=84)
Let's program and play (unit? unit=78&lesson=85)
Dobble Game - Spot the similarity 01 (unit? unit=78&lesson=86)
Dobble Game - Spot the similarity 02 (unit? unit=78&lesson=87)
Dobble Game - Spot the

2) What would the magic constant be for a magic square of size 7, given that all elements in the square are distinct numbers from 1 to 49? ***1 point***

- 260
- 111
- 175
- 165

Yes, the answer is correct.
Score: 1
Accepted Answers:
175

3) Does transposing any magic square change the sum across some rows/columns/diagonal? ***1 point***

- Yes
- No

Yes, the answer is correct.
Score: 1
Accepted Answers:
No

4) Which of the following are valid magic squares ? ***1 point***

$$\begin{bmatrix} 10 & 3 & 13 & 8 \\ 5 & 16 & 2 & 11 \\ 4 & 9 & 7 & 14 \\ 15 & 6 & 12 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 20-e & 6-e & 26-e & 16-e \\ 10-e & 32-e & 4-e & 22-e \\ 8-e & 18-e & 14-e & 28-2e \\ 30-e & 12-e & 24-e & 2-e \end{bmatrix}$$

$$\begin{bmatrix} 1 & 14 & 4 & 15 \\ 8 & 11 & 5 & 10 \\ 13 & 2 & 16 & 3 \\ 12 & 7 & 9 & 6 \end{bmatrix}$$

$$\begin{bmatrix} \pi & 14\pi & 4\pi & 15\pi \\ 8\pi & 11\pi & 5\pi & 10\pi \\ 13\pi & 2\pi & 16\pi & 3\pi \\ 12\pi & 7\pi & 9\pi & 6\pi \end{bmatrix}$$

No, the answer is incorrect.

similarity 03 (unit?unit=78&lesson=88)
Dobble Game - Spot the similarity 04 (unit?unit=78&lesson=89)
What is your date of birth? (unit?unit=78&lesson=90)
Birthday Paradox - Find your twin 01 (unit?unit=78&lesson=91)
Birthday Paradox - Find your twin 02 (unit?unit=78&lesson=92)
Birthday Paradox - Find your twin 03 (unit?unit=78&lesson=93)
Birthday Paradox - Find your twin 04 (unit?unit=78&lesson=94)
Birthday Paradox - Find your twin 05 (unit?unit=78&lesson=95)
What's your favourite movie? (unit?

Score: 0

Accepted Answers:

$$\begin{bmatrix} 10 & 3 & 13 & 8 \\ 5 & 16 & 2 & 11 \\ 4 & 9 & 7 & 14 \\ 15 & 6 & 12 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 14 & 4 & 15 \\ 8 & 11 & 5 & 10 \\ 13 & 2 & 16 & 3 \\ 12 & 7 & 9 & 6 \end{bmatrix}$$

$$\begin{bmatrix} \pi & 14\pi & 4\pi & 15\pi \\ 8\pi & 11\pi & 5\pi & 10\pi \\ 13\pi & 2\pi & 16\pi & 3\pi \\ 12\pi & 7\pi & 9\pi & 6\pi \end{bmatrix}$$

5) What is the minimum number of people required to ensure that at least three of them share the same 30-minute birth interval? The intervals start from 12:00 AM and each interval lasts for half an hour.

97

Yes, the answer is correct.
Score: 1

Accepted Answers:
(Type: Numeric) 97

1 point

6) Calculate the magic constant for 5x5 square, where all elements are distinct numbers from 1 to 25, is it same as the magic constant for Ramanujan's magic square ?

If yes, enter 0, else enter the absolute difference between the two.
Hint: Search the about Ramanujan's magic square.

50

No, the answer is incorrect.
Score: 0

Accepted Answers:
(Type: Numeric) 74

1 point

7) What task does **mystery1()** perform? ***1 point***

unit=78&lesson=96)

Guess the Movie Name 01 (unit?unit=78&lesson=97)

Guess the Movie Name 02 (unit?unit=78&lesson=98)

Guess the Movie Name 03 (unit?unit=78&lesson=99)

Guess the Movie Name 04 (unit?unit=78&lesson=100)

Guess the Movie Name 05 (unit?unit=78&lesson=101)

Guess the Movie Name 06 (unit?unit=78&lesson=102)

Week 4 Feedback Form: The Joy of Computing using Python (unit?unit=78&lesson=103)

Quiz: Week 4 : Assignment 4 (assessment?name=515)

Week 4: Programming Assignment 1

```python
def mystery1(number):
  list1 = []
  for i in range(1, number):
    if number % i == 0:
      list1.append(i)
  return list1

def mystery2(n1,n2):
  flag = False
  list2 = []
  for i in mystery1(n1):
    for j in mystery1(n2):
      if i == j:
        flag = True
        list2.append(i)
  if len(list2) > 1:
    print("Completed")
```

- Calculate factorial of number n.
- Calculate factors of number n.
- Calculate factors of number n+1 excluding n.
- Calculate factors of number n excluding n.

Yes, the answer is correct.
Score: 1

Accepted Answers:
Calculate factors of number n excluding n.

8) For what **n1, n2** will the variable **flag** inside **mystery2()** be equal to **True**? ***1 point***

- [x] 1,2
- [x] 2,3
- [x] 3,4
- [ ] 0,0

No, the answer is incorrect.
Score: 0

Accepted Answers:
2,3
3,4

9) If all possible pairs of prime numbers between 0 and 10, are given to n1 and n2, for how many pairs would 2 print ”Completed” ? ***1 point***

- It will print ”Completed” only for pairs (2,3)(7,2),(2,5), and for the remaining it would not print ”Completed”.

(/noc25_cs69/progassignment?name=518)

Week 4: Programming Assignment 2 (/noc25_cs69/progassignment?name=516)

Week 4: Programming Assignment 3 (/noc25_cs69/progassignment?name=517)

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- It will print ”Completed” only for pair (2,3), and for the remaining other pairs of primes it would not print ”Completed”.
- It will print ”Completed” for all pairs of primes between 0 and 20.
- It will not print ”Completed” for any pair.

Yes, the answer is correct.
Score: 1

Accepted Answers:
It will not print ”Completed” for any pair.

10) If numbers of pairs of primes which result in mystery2 to print ”Completed” are lesser than 1, Can we edit the code in **mystery2()** so that ”Completed” is always printed for any pair of primes ? ***1 point***

- Yes, we can change the logic for setting **flag** variable to False.
- No, it is logically not possible.
- Yes, we can change/decrease the threshold for length of **list2** in the last **if** block.
- Yes, we can change the initial value of flag to False, instead of True.

Yes, the answer is correct.
Score: 1

Accepted Answers:
*Yes, we can change/decrease the threshold for length of **list2** in the last **if** block.*